

The Exponent Rules

Exponent: A superscript.

Base: A number that has a superscript.

Power: A numerical expression with a base and a superscript.

Example



Exponent Rules Learned in Pre-Algebra

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

I ① $x^6 \cdot x^4 = x^{6+4} = \boxed{x^{10}}$

I ② $x^2 \cdot x^3 = \boxed{x^5}$

I ③ $x^4 \cdot x^8 \cdot x^5 = \boxed{x^{17}}$

I ④ $x^9 \cdot x^2 = \boxed{x^{11}}$

I ⑤ $x^{10} \cdot x^3 = \boxed{x^{13}}$

I ⑥ $x^8 \cdot x^7 \cdot x^4 = \boxed{x^{19}}$

II ⑦ $\frac{x^6}{x^4} = x^{6-4} = \boxed{x^2}$

II ⑧ $\frac{x^7}{x^3} = \boxed{x^4}$

II ⑨ $\frac{x^5}{x^2} = \boxed{x^3}$

II ⑩ $\frac{x^8}{x^6} = \boxed{x^2}$

Exponent Rules

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Examples Demonstrating the Reasoning behind Rule I and II.

Example: $x^3 \cdot x^4 = (x \cdot x \cdot x)(x \cdot x \cdot x \cdot x)$ No need for the parentheses.
 $= \underset{1}{x} \cdot \underset{2}{x} \cdot \underset{3}{x} \cdot \underset{4}{x} \cdot \underset{5}{x} \cdot \underset{6}{x} \cdot \underset{7}{x}$
 $= x^7$

Conclusion: Rule I) $x^a \cdot x^b = x^{a+b}$

Example: $\frac{x^5}{x^3} = \frac{\underset{\text{yellow}}{x \cdot x \cdot x} \cdot \underset{\text{blue}}{x \cdot x}}{\underset{\text{yellow}}{x \cdot x \cdot x}} = \underset{\text{blue}}{x \cdot x} = x^2$ The yellow x's cancel out.

Conclusion: Rule II) $\frac{x^a}{x^b} = x^{a-b}$

New Exponent Rules

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

III ⑪ $(3x)^2 = 3^2 \cdot x^2 = 9 \cdot x^2 = \boxed{9x^2}$

III ⑫ $(-2y)^3 = (-2)^3 \cdot y^3 = \boxed{-8y^3}$

III ⑬ $(4ab)^2 = 4^2 \cdot a^2 \cdot b^2 = \boxed{16a^2b^2}$

III ⑭ $(5x)^2 = 5^2 \cdot x^2 = \boxed{25x^2}$

III ⑮ $(-3y)^3 = (-3)^3 \cdot y^3 = \boxed{-27y^3}$

III ⑯ $(2yb)^4 = 2^4 \cdot y^4 \cdot b^4 = \boxed{16y^4b^4}$

An Example Demonstrating the Reasoning behind Rule III.

$$(xy)^3 = (xy)(xy)(xy) = \underset{\uparrow}{(x \cdot x \cdot x)(y \cdot y \cdot y)} = x^3 y^3$$

Commutative Rule of Multiplication

Conclusion: Rule III) $(xy)^a = x^a \cdot y^a$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

IV ①7 $\left(\frac{x}{3}\right)^2 = \frac{x^2}{3^2} = \boxed{\frac{x^2}{9}}$

IV ①8 $\left(\frac{4}{y}\right)^3 = \frac{4^3}{y^3} = \boxed{\frac{64}{y^3}}$

IV ①9 $\left(\frac{a}{-3}\right)^4 = \frac{a^4}{(-3)^4} = \boxed{\frac{a^4}{81}}$

IV ②0 $\left(\frac{5}{b}\right)^2 = \frac{5^2}{b^2} = \boxed{\frac{25}{b^2}}$

IV ②1 $\left(\frac{x}{2}\right)^3 = \frac{x^3}{2^3} = \boxed{\frac{x^3}{8}}$

IV ②2 $\left(\frac{-1}{x}\right)^3 = \frac{(-1)^3}{x^3} = \boxed{\frac{-1}{x^3}}$

An Example Demonstrating the Reasoning behind Rule IV.

$$\left(\frac{x}{y}\right)^3 = \left(\frac{x}{y}\right)\left(\frac{x}{y}\right)\left(\frac{x}{y}\right) = \frac{x \cdot x \cdot x}{y \cdot y \cdot y} = \frac{x^3}{y^3}$$

Conclusion: Rule IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

V ②3 $(x^2)^5 = x^{2 \cdot 5} = \boxed{x^{10}}$

V ②4 $(b^3)^3 = \boxed{b^9}$

V ②5 $(y^4)^3 = \boxed{y^{12}}$

V ②6 $(a^5)^6 = \boxed{a^{30}}$

An Example Demonstrating the Reasoning behind Rule V.

$$(x^2)^5 = x^2 \cdot x^2 \cdot x^2 \cdot x^2 \cdot x^2 = x^{10}$$

Rule I) $x^a \cdot x^b = x^{a+b}$

Conclusion: Rule V) $(x^a)^b = x^{ab}$

More Practice

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

IV ②7 $\left(\frac{3}{a}\right)^2 = \frac{3^2}{a^2} = \boxed{\frac{9}{a^2}}$

I ②8 $c^6 \cdot c^2 = \boxed{c^8}$

$$\text{III } (29) (4b)^3 = 4^3 b^3 = \boxed{64b^3}$$

$$\text{V } (30) (y^3)^2 = \boxed{y^6}$$

$$\text{II } (31) \frac{b^7}{b^2} = \boxed{b^5}$$

$$\text{V } (32) (b^4)^2 = \boxed{b^8}$$

$$\text{I } (33) x^6 \cdot x^3 = \boxed{x^9}$$

$$\text{III } (34) (6y)^2 = 6^2 y^2 = \boxed{36y^2}$$

$$\text{IV } (35) \left(\frac{x}{4}\right)^2 = \frac{x^2}{4^2} = \boxed{\frac{x^2}{16}}$$

$$\text{II } (36) \frac{a^6}{a^2} = \boxed{a^4}$$

Strange New Exponent Rules

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

$$\text{VII } (37) (-9)^0 = \boxed{1}$$

$$\text{VII } (38) b^0 = \boxed{1}$$

$$\text{VII } (39) (3xy^2)^0 = \boxed{1}$$

$$\text{VI } (40) 2^1 = \boxed{2}$$

$$\text{VI } (41) a^1 = \boxed{a}$$

$$\text{VI } (42) (5a)^1 = \boxed{5a}$$

$$\text{VII } (43) 3^0 = \boxed{1}$$

$$\text{VII } (44) c^0 = \boxed{1}$$

$$\text{VII } (45) (-7ab^3c^2)^0 = \boxed{1}$$

$$\text{VI } (46) 23^1 = \boxed{23}$$

$$\text{VI } (47) (-8)^1 = \boxed{-8}$$

$$\text{VI } (48) (4x^3y^5z)^1 = \boxed{4x^3y^5z}$$

Examples Demonstrating the Reasoning behind Rule VI and VII

Example: $\frac{x^4}{x^4} = 1$

$x \neq 0$ (green arrow pointing to x^4)

A number divided by itself is 1. (pink arrow pointing to $= 1$)

Rule II) $\frac{x^a}{x^b} = x^{a-b}$

$x^{4-4} = 1$

$x^0 = 1$

Conclusion: Rule VII) $x^0 = 1$

Example: $x^5 \cdot x = x^6$

Example: $x^a \cdot x = x^{a+1}$

Rule I) $x^a \cdot x^b = x^{a+b}$ Conclusion: $b=1$ & VI) $x = x^1$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

VIII (49) $3^{-2} = \frac{1}{3^2} = \boxed{\frac{1}{9}}$

VIII (50) $2^{-4} = \frac{1}{2^4} = \boxed{\frac{1}{16}}$

VIII (51) $4^{-3} = \frac{1}{4^3} = \boxed{\frac{1}{64}}$

VIII (52) $x^{-8} = \boxed{\frac{1}{x^8}}$

VIII (53) $2^{-3} = \frac{1}{2^3} = \boxed{\frac{1}{8}}$

VIII (54) $4^{-2} = \frac{1}{4^2} = \boxed{\frac{1}{16}}$

VIII (55) $3^{-3} = \frac{1}{3^3} = \boxed{\frac{1}{27}}$

VIII (56) $y^{-5} = \boxed{\frac{1}{y^5}}$

An Example Demonstrating the Reasoning behind Rule VIII.

Example: $\frac{x^3}{x^5} = \frac{1}{x^2}$

Rule II) $\frac{x^a}{x^b} = x^{a-b}$ $\rightarrow x^{3-5} = \frac{1}{x^2}$

$x^{-2} = \frac{1}{x^2}$

Conclusion: Rule VIII) $x^{-a} = \frac{1}{x^a}$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

IX (57) $9^{\frac{1}{2}} = \sqrt{9} = \boxed{3}$

IX (58) $8^{\frac{1}{3}} = \sqrt[3]{8} = \boxed{2}$

IX (59) $32^{\frac{1}{5}} = \sqrt[5]{32} = \boxed{2}$

IX (60) $b^{\frac{1}{6}} = \boxed{\sqrt[6]{b}}$

IX (61) $x^{\frac{1}{8}} = \boxed{\sqrt[8]{x}}$

$$\text{IX } ⑥② \ 49^{\frac{1}{2}} = \sqrt{49} = \boxed{7}$$

$$\text{IX } ⑥③ \ 27^{\frac{1}{3}} = \sqrt[3]{27} = \boxed{3}$$

$$\text{IX } ⑥④ \ 16^{\frac{1}{4}} = \sqrt[4]{16} = \boxed{2}$$

$$\text{IX } ⑥⑤ \ a^{\frac{1}{5}} = \sqrt[5]{a}$$

$$\text{IX } ⑥⑥ \ y^{\frac{1}{7}} = \sqrt[7]{y}$$

The Reasoning behind Rule IX.

Raise both sides
of the equation
to the n^{th} power.

$$\text{If } x^c = \sqrt[n]{x}$$

then

$$(x^c)^n = (\sqrt[n]{x})^n$$

$$\text{Rule V) } (x^a)^b = x^{ab}$$

$$x^{cn} = x$$

$$x^{cn} = x^1$$

Rule VI)

$$x = x^1$$

Because the bases
are the same.

$$cn = 1$$

Plug $\frac{1}{n}$ into
the original
equation.

$$c = \frac{1}{n}$$

$$\text{Conclusion: Rule IX) } x^{\frac{1}{n}} = \sqrt[n]{x}$$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

$$\text{X } ⑥⑦ \ 25^{\frac{3}{2}} = (\sqrt{25})^3 = 5^3 = \boxed{125}$$

$$\text{X } ⑥⑧ \ (-8)^{\frac{2}{3}} = (\sqrt[3]{-8})^2 = (-2)^2 = \boxed{4}$$

$$\text{X } ⑥⑨ \ 16^{\frac{3}{4}} = (\sqrt[4]{16})^3 = (2)^3 = \boxed{8}$$

$$\text{X } ⑦⑩ \ a^{\frac{5}{6}} = (\sqrt[6]{a})^5$$

$$\text{X } ⑦⑪ \ x^{\frac{10}{7}} = (\sqrt[7]{x})^{10}$$

$$\text{X } ⑦⑫ \ 4^{\frac{3}{2}} = (\sqrt{4})^3 = (2)^3 = \boxed{8}$$

$$\text{X } ⑦⑬ \ (-27)^{\frac{2}{3}} = (\sqrt[3]{-27})^2 = (-3)^2 = \boxed{9}$$

$$\text{X } ⑦⑭ \ 81^{\frac{3}{4}} = (\sqrt[4]{81})^3 = (3)^3 = \boxed{27}$$

$$\text{X } (75) \ b^{\frac{8}{11}} = \boxed{(\sqrt[11]{b})^8}$$

$$\text{X } (76) \ c^{\frac{3}{5}} = \boxed{(\sqrt[5]{c})^3}$$

The Reasoning behind Rule X.

Start with Rule IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$

Raise both sides of the equation to the b^{th} power.

$$(x^{\frac{1}{a}})^b = (\sqrt[a]{x})^b$$

$$\text{Rule II) } (x^c)^h = x^{ch} \rightarrow x^{\frac{1}{a} \cdot b} = (\sqrt[a]{x})^b$$

$$\text{Conclusion: Rule X) } x^{\frac{b}{a}} = (\sqrt[a]{x})^b$$

More Practice (Rules VII-X)

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

$$\text{VII } (77) \ 4^1 = \boxed{4}$$

$$\text{IX } (78) \ 64^{\frac{1}{2}} = \sqrt{64} = \boxed{8}$$

$$\text{VIII } (79) \ 6^{-2} = \frac{1}{6^2} = \boxed{\frac{1}{36}}$$

$$\text{VII } (80) \ 12^0 = \boxed{1}$$

$$\text{VI } (81) \ 57^1 = \boxed{57}$$

$$\text{X } (82) \ 125^{\frac{2}{3}} = (\sqrt[3]{125})^2 = (5)^2 = \boxed{25}$$

$$\text{VII } (83) \ 100^0 = \boxed{1}$$

$$\text{IX } (84) \ x^{\frac{1}{9}} = \boxed{\sqrt[9]{x}}$$

$$\text{VIII } (85) \ 10^{-3} = \frac{1}{10^3} = \boxed{\frac{1}{1000}}$$

$$\text{X } (86) \ 9^{\frac{3}{2}} = (\sqrt{9})^3 = (3)^3 = \boxed{27}$$

$$\text{IX } (87) \ 16^{\frac{1}{4}} = \sqrt[4]{16} = \boxed{2}$$

$$\text{VIII } (88) \ b^{-2} = \boxed{\frac{1}{b^2}}$$

$$\text{VII } (89) \ a^0 = \boxed{1}$$

$$\text{VI } (90) \ y^1 = \boxed{y}$$

$$\text{VII } (91) \ (-3xy^5)^0 = \boxed{1}$$

Rule X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

X 92) $\sqrt[3]{27^2} = (\sqrt[3]{27})^2 = (3)^2 = \boxed{9}$

X 93) $\sqrt{25^3} = (\sqrt{25})^3 = (5)^3 = \boxed{125}$

X 94) $\sqrt[4]{16^2} = (\sqrt[4]{16})^2 = (2)^2 = \boxed{4}$

X 95) $\sqrt[3]{8^4} = (\sqrt[3]{8})^4 = (2)^4 = \boxed{16}$

X 96) $\sqrt{9^3} = (\sqrt{9})^3 = 3^3 = \boxed{27}$

X 97) $\sqrt[4]{81^3} = (\sqrt[4]{81})^3 = 3^3 = \boxed{27}$

X 98) $\sqrt[3]{64^2} = (\sqrt[3]{64})^2 = (4)^2 = \boxed{16}$

X 99) $\sqrt{16^3} = (\sqrt{16})^3 = 4^3 = \boxed{64}$

The Reasoning behind the Variation of Rule X.

$$x^{\frac{b}{a}} = x^{\frac{1}{a} \cdot b} = (x^{\frac{1}{a}})^b = (\sqrt[a]{x})^b$$

Commutative Rule of Multiplication.

Rule II) $x^{c \cdot h} = (x^c)^h$

$$= x^{b \cdot \frac{1}{a}} = (x^b)^{\frac{1}{a}} = \sqrt[a]{x^b}$$

Rule IX) $x^{\frac{1}{n}} = \sqrt[n]{x}$

Conclusion: Rule X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$

More Practice (Rules I-X)

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

VIII 100) $2^{-5} = \frac{1}{2^5} = \boxed{\frac{1}{32}}$

V 101) $(b^3)^6 = b^{3 \cdot 6} = \boxed{b^{18}}$

VII 102) $8^0 = \boxed{1}$

I 103) $x^9 \cdot x^8 = \boxed{x^{17}}$

$$\text{III } (104) (7a)^2 = 7^2 a^2 = \boxed{49a^2}$$

$$\text{'X } \star (105) \sqrt[5]{8^5} = (\sqrt[5]{8})^5 = (2)^5 = \boxed{32}$$

$$\text{VI } (106) 15^1 = \boxed{15}$$

$$\text{II } (107) \frac{y^7}{y^4} = y^{7-4} = \boxed{y^3}$$

$$\text{X } (108) x^{\frac{6}{7}} = \boxed{(\sqrt[7]{x})^6}$$

$$\text{IV } (109) \left(\frac{b}{-4}\right)^2 = \frac{b^2}{(-4)^2} = \boxed{\frac{b^2}{16}}$$

$$\text{VI } (110) c^1 = \boxed{c}$$

$$\text{V } (111) (a^2)^{10} = a^{2 \cdot 10} = \boxed{a^{20}}$$

$$\text{VIII } (112) 9^{-2} = \frac{1}{9^2} = \boxed{\frac{1}{81}}$$

$$\text{'X } \star (113) \sqrt[4]{9^4} = (\sqrt[4]{9})^4 = 3^4 = \boxed{81}$$

$$\text{VII } (114) y^0 = \boxed{1}$$

$$\text{I } (115) a^6 \cdot a^6 = \boxed{a^{12}}$$

$$\text{IX } (116) b^{\frac{1}{4}} = \boxed{\sqrt[4]{b}}$$

$$\text{III } (117) (-4xy)^3 = (-4)^3 x^3 y^3 = \boxed{-64x^3y^3}$$

$$\text{II } (118) \frac{a^{10}}{a^4} = a^{10-4} = \boxed{a^6}$$

$$\text{X } (119) 8^{\frac{5}{3}} = (\sqrt[3]{8})^5 = (2)^5 = \boxed{32}$$

$$\text{VII } (120) (3a^{10}b^7)^1 = \boxed{3a^{10}b^7}$$

$$\text{IX } (121) 4^{\frac{1}{2}} = \sqrt{4} = \boxed{2}$$

$$\text{IV } (122) \left(\frac{2}{a}\right)^3 = \frac{2^3}{a^3} = \boxed{\frac{8}{a^3}}$$